

Uncountable Auerbach Systems in the Koszmider–Wark Separable-Range Renormings

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Status

This packet gives a candidate full answer to the Auerbach-system question raised by Koszmider and Wark in [1]. After Theorem 3, they consider renormings

$$\|x\|_T = \|x\|_1 + \|Tx\| \quad (x \in \ell_1([0, 1])),$$

where $T : \ell_1([0, 1]) \rightarrow X$ is injective and has separable range, and write that they do not know whether these spaces admit an uncountable Auerbach system. The theorem below proves that they do. Injectivity of T is not needed for this conclusion.

Source Evidence

The source statement appears on page 3 of [1], immediately after Theorem 3.

~~many infinite equilateral sets (see Remark 10). Theorem 3.~~

Theorem 3. *Suppose that $\mathcal{X}$ is a Banach space and $T : \ell_1([0, 1]) \rightarrow \mathcal{X}$ is injective and has separable range and $r > 0$. Then $(\ell_1([0, 1]), \|\cdot\|_T)$ has the following property: Any r -separated subset $\mathcal{Y} \subseteq \ell_1([0, 1])$ of regular uncountable cardinality has a subset $\mathcal{Z} \subseteq \mathcal{Y}$ of the same cardinality which is $(r+)$ -separated. In particular $(\ell_1([0, 1]), \|\cdot\|_T)$ does not admit any uncountable equilateral set.*

The above property for renormings induced by T injective compact and with strictly convex range is a consequence of Theorem 1 and some partition calculus results (see Corollary 7). Also this property is much stronger than not having uncountable equilateral sets (see Remark 11).

A close link between separated subsets in the sphere and Auerbach bases was demonstrated in [8]. In fact Godun’s renorming of $\ell_1([0, 1])$ was designed to prove that the space has no fundamental Auerbach system ([6]). Nevertheless, we do not know if our spaces admits an uncountable Auerbach system.

Let us also remark that considering Banach spaces without large equilateral

Definitions

An Auerbach system in a Banach space Z is a family

$$(z_i, z_i^*)_{i \in I} \subseteq Z \times Z^*$$

such that $\|z_i\| = \|z_i^*\| = 1$, $z_i^*(z_i) = 1$, and $z_i^*(z_j) = 0$ whenever $i \neq j$.

Proof Intuition

The separable range of T forces many coordinate vectors of $\ell_1([0, 1])$ to have very close T -images. Taking disjoint pairs of such coordinates gives uncountably many block vectors $u_i = e_{a_i} - e_{b_i}$ with uniformly small Tu_i . On the closed span of these disjoint blocks, the perturbation term $\|T\cdot\|$ is dominated by the original ℓ_1 -norm. Consequently each block vector is exactly as far from the span of all the other blocks as it is from the origin. The Hahn–Banach theorem turns this distance-attainment statement into the desired norm-one biorthogonal functionals.

Theorem 1. *Let Γ be an uncountable set, let Y be a Banach space, and let $T : \ell_1(\Gamma) \rightarrow Y$ be a bounded linear operator with separable range. Define an equivalent norm on $\ell_1(\Gamma)$ by*

$$\|x\|_T = \|x\|_1 + \|Tx\|_Y.$$

Then $(\ell_1(\Gamma), \|\cdot\|_T)$ contains an uncountable Auerbach system.

Proof. For $\gamma \in \Gamma$, denote by e_γ the canonical unit vector of $\ell_1(\Gamma)$. Since the range of T is separable, the set $\{Te_\gamma : \gamma \in \Gamma\}$ is contained in a separable metric space. Cover this separable space by countably many open balls of radius $1/2$. One of these balls contains Te_γ for uncountably many γ 's. Let $\Gamma_0 \subseteq \Gamma$ be an uncountable set such that

$$\|Te_\gamma - Te_\eta\|_Y \leq 1 \quad (\gamma, \eta \in \Gamma_0).$$

Choose pairwise disjoint two-point subsets $\{a_i, b_i\} \subseteq \Gamma_0$, indexed by an uncountable set I . Put

$$u_i = e_{a_i} - e_{b_i} \quad (i \in I).$$

Then $\|Tu_i\|_Y \leq 1$ for every $i \in I$. Let

$$E = \overline{\text{span}}\{u_i : i \in I\}.$$

Because the supports of the u_i 's are disjoint, every $y \in E$ has a unique representation $y = \sum_{i \in I} c_i u_i$, with $\sum_i |c_i| < \infty$, and

$$\|y\|_1 = 2 \sum_i |c_i|.$$

Moreover,

$$\|Ty\|_Y \leq \sum_i |c_i| \|Tu_i\|_Y \leq \sum_i |c_i| = \frac{1}{2} \|y\|_1 \leq \|y\|_T.$$

Fix $i \in I$, and let

$$E_i = \overline{\text{span}}\{u_j : j \in I, j \neq i\}.$$

For $y \in E_i$, the supports of u_i and y are disjoint, hence

$$\|u_i - y\|_1 = \|u_i\|_1 + \|y\|_1.$$

Using the reverse triangle inequality and the estimate just proved for $y \in E$, we get

$$\begin{aligned} \|u_i - y\|_T &= \|u_i - y\|_1 + \|Tu_i - Ty\|_Y \\ &\geq \|u_i\|_1 + \|y\|_1 + \|Tu_i\|_Y - \|Ty\|_Y \\ &\geq \|u_i\|_1 + \|Tu_i\|_Y \\ &= \|u_i\|_T. \end{aligned}$$

Thus $\text{dist}_{\|\cdot\|_T}(u_i, E_i) = \|u_i\|_T$.

Define a functional g_i on $\text{span}(E_i \cup \{u_i\}) = E$ by

$$g_i(\alpha u_i + y) = \alpha \|u_i\|_T \quad (\alpha \in \mathbb{R}, y \in E_i).$$

The distance estimate shows that $\|g_i\|_{E^*} = 1$, and clearly $g_i(u_i) = \|u_i\|_T$ while $g_i(u_j) = 0$ for $j \neq i$. By the Hahn–Banach theorem, extend g_i to a norm-one functional $\tilde{g}_i \in (\ell_1(\Gamma), \|\cdot\|_T)^*$.

Finally set $x_i = u_i / \|u_i\|_T$ and $x_i^* = \tilde{g}_i$. Then

$$\|x_i\|_T = \|x_i^*\| = 1, \quad x_i^*(x_i) = 1, \quad x_i^*(x_j) = 0 \quad (i \neq j).$$

Since I is uncountable, $(x_i, x_i^*)_{i \in I}$ is an uncountable Auerbach system. $\square$

Corollary 1. *Every Koszmider–Wark renorming from Theorem 3 of [1] admits an uncountable Auerbach system. In particular, the compact strictly convex examples from Theorem 1, which have no infinite equilateral set, admit uncountable Auerbach systems.*

Proof. Apply the theorem with $\Gamma = [0, 1]$. The hypotheses of Theorem 3 in [1] include separable range for T . The compact examples from Theorem 1 also have separable range, since compact subsets of Banach spaces are separable in the norm metric. $\square$

Verification and Novelty Notes

The local run indexes were searched for arXiv:2104.05330, the title, the phrases “uncountable Auerbach”, “infinite equilateral”, and “separable range”. The existing nearby packet for arXiv:1503.06356/arXiv:2104.05335 concerns the separate absolute-example question for uncountable equilateral sets, not this Auerbach-system question.

A web search for the source title together with “Auerbach system”, for “2104.05330 Auerbach system”, and for close variants found the later Koszmider paper [2]. That paper references the Koszmider–Wark examples, but the search did not reveal an explicit answer to the present separable-range Auerbach question. The proof above is self-contained apart from standard Hahn–Banach separation.

Limitations

This packet answers the Auerbach question for the spaces of Theorem 3 in [1]. It does not address the separate open questions in [1] about possible densities above the continuum, about all equivalent renormings admitting uncountable equilateral sets, or about $C(K)$ -specific absolute examples.

Human Review Recommendation

The main point to check is the estimate

$$\|Ty\|_Y \leq \|y\|_1 \quad (y \in E),$$

obtained from the disjoint block construction, and its use in proving $\text{dist}(u_i, E_i) = \|u_i\|_T$. Once that estimate is accepted, the Hahn–Banach construction of the Auerbach functionals is routine.

References

- [1] P. Koszmider and H. Wark, *Large Banach spaces with no infinite equilateral sets*, arXiv:2104.05330, 2021.
- [2] P. Koszmider, *On Ramsey-type properties of the distance in nonseparable spheres*, arXiv:2308.07668, 2023.